

PART A (5×2=10 Marks)

Answer ALL questions. Each question carries 2 marks.

What are the components of an ecosystem?

What are biodiversity hotspots?

What is the role of the Forest Conservation Act, 1980?

Mention two environmental benefits of wind energy.

What is the significance of sustainable development?

Expand OHSMS and define it.

PART B (2×16=32 Marks)

Answer ANY TWO questions. Each question carries 16 marks.

6(a) Explain the structure and functions of an ecosystem. (8)

Describe different types of ecological pyramids. (8)

OR

6(b) Discuss the different types of biodiversity and their importance. (8)

What are the causes of biodiversity loss? (8)

7(a) Explain how and why electronic waste (e-waste) has become a growing solid waste problem OR WATER POLLUTION

OR

7(b) What are the salient features of environmental (protection) Act 1986.(8)

Discuss the methods of disposal of municipal solid waste management.(8)

PART C (1×8=8 Marks)

Answer ANY ONE question. Each question carries 8 marks.

8(a) Explain the principle and various steps involved in the energy management.(8)

OR

8(b) Write a note on (i) tidal energy (ii) OTE (iii) GTE(8)

ANSWERS>>>

Structure and Functions of an Ecosystem (8 Marks)

An **ecosystem** is a system where living organisms interact with each other and their physical environment. It consists of **biotic (living)** and **abiotic (non-living)** components.

Structure of an Ecosystem:

1. **Biotic Components (Living Organisms)**

- **Producers (Autotrophs):** Plants, algae, and some bacteria that produce energy via photosynthesis.
- **Consumers (Heterotrophs):** Organisms that depend on others for food. These include:
 - **Primary Consumers:** Herbivores (e.g., deer, rabbits).
 - **Secondary Consumers:** Carnivores (e.g., snakes, foxes).
 - **Tertiary Consumers:** Top predators (e.g., lions, eagles).
- **Decomposers (Saprotrophs):** Bacteria and fungi that break down organic matter into nutrients.

1. **Abiotic Components (Non-Living Factors)**

- **Climatic Factors:** Light, temperature, humidity.
- **Edaphic Factors:** Soil type, minerals, pH level.

- **Chemical Substances:** Oxygen, nitrogen, carbon dioxide, water.

Functions of an Ecosystem:

1. **Energy Flow:**
 - Energy moves in **one direction** from the Sun → Producers → Consumers → Decomposers.
 - The **10% law** states that only **10% of energy** is passed to the next trophic level.
1. **Nutrient Cycling:**
 - Essential elements like **carbon, nitrogen, phosphorus** are cycled through **biogeochemical cycles**.
1. **Ecological Balance:**
 - Predators keep prey populations in check, maintaining balance.
1. **Homeostasis:**
 - Ecosystems adjust to external changes through feedback mechanisms.
1. **Pollution Control:**
 - Plants and microorganisms help clean air, water, and soil.

Different Types of Ecological Pyramids (8 Marks)

Ecological pyramids are graphical representations of energy flow, biomass, or number of organisms at each trophic level.

1. Pyramid of Numbers:

- Shows the **number of organisms** at each trophic level.
- **Example:** In a grassland ecosystem, grass is more numerous than herbivores (deer), which are more than carnivores (lions).
- **Inverted in some cases:** A single tree can support many birds and insects.

2. Pyramid of Biomass:

- Represents the **total dry weight of organisms** at each level.
- Usually **upright**, as biomass decreases up the food chain.
- **Example:** In a forest, the biomass of trees (producers) is more than herbivores (deer) and even less for carnivores (tigers).
- **Inverted in aquatic ecosystems** since phytoplankton reproduce quickly but have lower biomass than zooplankton.

3. Pyramid of Energy:

- Shows **energy flow** at each level, always **upright**.
- Follows the **10% energy transfer rule**—energy decreases as we go up the pyramid.
- **Example:**
- Sunlight (100%) → Plants (10%) → Herbivores (1%) → Carnivores (0.1%).

Diagram Representation:

- Grassland Pyramid (Numbers) → Upright
- Tree Pyramid (Numbers) → Inverted
- Energy Pyramid → Always Upright

OR

(b) Different Types of Biodiversity and Their Importance (8 Marks)

Biodiversity refers to the **variety of life forms** on Earth, including plants, animals, and microorganisms, along with their ecosystems. It is classified into three main types:

1. Genetic Biodiversity:

- **Definition:** Variation in genes within a species.
- **Example:** Different breeds of dogs, varieties of rice, and mango species.
- **Importance:**
- Helps species **adapt** to environmental changes.
- Increases **disease resistance** in crops and livestock.

2. Species Biodiversity:

- **Definition:** Variety of species in an ecosystem.
- **Example:** Rainforests have diverse species like tigers, elephants, and parrots.
- **Importance:**
- **Ecosystem stability** – More species ensure balance.
- **Medicinal value** – Many species provide natural medicines.

3. Ecosystem Biodiversity:

- **Definition:** Variety of ecosystems like forests, deserts, wetlands, and oceans.
- **Example:** Tropical rainforests, coral reefs, and mangroves.
- **Importance:**
- Provides **natural resources** like food, water, and oxygen.
- Supports **climate regulation** and disaster prevention.

Causes of Biodiversity Loss (8 Marks)

1. Deforestation:

- Clearing forests for agriculture, urbanization, and industries.
- Example: Amazon Rainforest deforestation reduces habitat for species.

1. Pollution:

- Air, water, and soil pollution harm plant and animal life.
- Example: **Oil spills** kill marine biodiversity.
- 1. **Climate Change:**
 - Rising temperatures, droughts, and floods affect species survival.
 - Example: **Coral bleaching** due to increased ocean temperatures.
- 1. **Overexploitation:**
 - **Excessive hunting, fishing, and logging** reduce species populations.
 - Example: **Poaching of tigers** for fur and bones.
- 1. **Invasive Species:**
 - Non-native species disrupt ecosystems and compete with native species.
 - Example: **Water hyacinth** spreads aggressively in lakes, choking aquatic life.
- 1. **Habitat Fragmentation:**
 - Construction of roads, dams, and cities **breaks ecosystems**, reducing wildlife movement.
 - Example: **Tiger corridors** are shrinking due to human settlements.
- 1. **Agricultural Expansion:**
 - **Monoculture farming** reduces genetic diversity and soil health.
 - Example: **Palm oil plantations** in Malaysia threaten orangutans.
- 1. **Illegal Wildlife Trade:**
 - Smuggling of animals and plants for **pet trade, medicine, and luxury goods**.
 - Example: **Ivory poaching** of elephants.

Conclusion:

Biodiversity is **essential for ecological balance**, and its loss leads to **climate instability and resource depletion**. Strict **conservation efforts** like afforestation, pollution control, and wildlife protection laws are needed to prevent further decline.

How and Why Electronic Waste (E-Waste) Has Become a Growing Solid Waste Problem (8 Marks)

What is E-Waste?

Electronic waste (e-waste) refers to **discarded electrical and electronic devices** such as computers, mobile phones, televisions, and household appliances. It contains **hazardous materials** like lead, mercury, cadmium, and brominated flame retardants, making it a major **solid waste problem**.

Why E-Waste is a Growing Problem:

1. **Rapid Technological Advancement:**
 - **Frequent upgrades** and new models make older devices obsolete quickly.
 - Example: Smartphones and laptops are replaced every **2-3 years**.
1. **Increase in Consumer Demand:**
 - The global **rise in population and digital dependency** leads to massive e-waste generation.
 - Example: India is the **third-largest producer of e-waste**, after China and the USA.
1. **Lack of Proper Disposal Methods:**
 - Many developing countries **lack recycling infrastructure**, leading to **improper dumping**.
 - **Informal recycling** exposes workers to toxic chemicals.
1. **Toxic Components in E-Waste:**
 - Contains **hazardous materials** like lead (Pb), mercury (Hg), and cadmium (Cd), which pollute **soil and water**.
 - Example: Lead in circuit boards causes **brain and kidney damage**.
1. **Illegal Export of E-Waste:**
 - Developed nations **dump e-waste in developing countries** for cheap disposal.
 - Example: **Guiyu, China** is a major hub for unsafe e-waste processing.
1. **Environmental Impact:**
 - **Landfills and incineration** of e-waste release **toxic fumes** into the atmosphere.
 - **Water pollution** occurs when chemicals leach into groundwater.

How E-Waste Can Be Managed:

1. **E-Waste Recycling:**
 - Setting up **formal recycling plants** to extract valuable materials like **gold, silver, and copper**.
 - Example: Japan uses **urban mining** to recover metals from e-waste.
1. **Extended Producer Responsibility (EPR):**
 - Companies should **take back old electronics** and ensure proper disposal.
 - Example: Brands like **Apple and Dell** offer e-waste recycling programs.
1. **Strict E-Waste Management Laws:**
 - The **E-Waste (Management) Rules, 2016 (India)** regulate disposal and recycling.
 - Ban on **illegal dumping and improper disposal**.
1. **Public Awareness:**
 - Educating people about **safe disposal methods** and **buying durable products**.
 - Encouraging **repairing instead of replacing gadgets**.

Conclusion:

E-waste is a **fast-growing solid waste problem** due to **technological advancements and improper disposal**. Without effective **recycling, regulation, and awareness**, it will continue to **harm the environment and human health**. Sustainable management is essential for a **greener future**.

OR

Water Pollution

Causes, Effects, and Preventive Measures of Water Pollution (8 Marks)

Causes of Water Pollution:

1. **Industrial Discharge** – Factories release toxic chemicals and heavy metals into rivers.
2. **Domestic Waste** – Untreated sewage contaminates freshwater sources.
3. **Agricultural Runoff** – Pesticides and fertilizers enter water bodies, causing eutrophication.
4. **Plastic Pollution** – Plastic waste blocks drainage and harms marine life.
5. **Oil Spills** – Accidental spills from ships damage marine ecosystems.

Effects of Water Pollution:

1. **Health Hazards** – Contaminated water causes diseases like cholera and typhoid.
2. **Marine Life Destruction** – Oxygen depletion leads to fish kills.
3. **Eutrophication** – Excessive nutrients cause algal blooms, reducing oxygen levels.
4. **Groundwater Contamination** – Toxic chemicals seep into drinking water sources.
5. **Economic Losses** – Fishermen and tourism industries suffer.

Preventive Measures:

1. **Wastewater Treatment** – Proper sewage treatment before discharge.
2. **Ban on Plastic Waste** – Reduce single-use plastics near water bodies.
3. **Sustainable Farming** – Minimize pesticide and fertilizer use.
4. **Oil Spill Management** – Use containment booms and bioremediation techniques.
5. **Strict Industrial Regulations** – Monitor and control factory waste disposal.

Case Study on Water Pollution (8 Marks)

Case Study: Ganga River Pollution (India)

- **Cause:**
- **Industrial waste** from tanneries, textile, and chemical plants.

- **Sewage discharge** from over 100 towns along the river.
- **Religious practices** like idol immersion and cremation.
- **Effects:**
- **High levels of heavy metals** (mercury, lead) affecting aquatic life.
- **Health hazards** (waterborne diseases like dysentery and hepatitis).
- **Loss of biodiversity** due to oxygen depletion.
- **Preventive Measures Taken:**
- **Namami Gange Programme (2014)** – Launched by the Indian government to clean the river.
- **Effluent treatment plants (ETPs)** set up to treat industrial waste.
- **Public awareness campaigns** to promote eco-friendly practices.

OR

Salient Features of the Environmental (Protection) Act, 1986 (8 Marks)

The **Environmental (Protection) Act, 1986** was enacted by the **Government of India** after the **Bhopal Gas Tragedy (1984)** to provide a framework for environmental protection. It grants **sweeping powers** to the **central government** to regulate and prevent pollution.

Key Features of the Act:

1. **Wide Scope of Coverage:**
 - Covers **air, water, land, and all living organisms**.
 - Addresses **hazardous substances management**.
1. **Power to the Central Government:**
 - The central government can **set standards, issue directions, and take legal action** against polluters.
 - Can **close industries** that violate environmental laws.
1. **Prevention of Environmental Pollution:**
 - Prohibits **discharge of pollutants** beyond prescribed limits.
 - Mandates **environmental impact assessments (EIA)** for major projects.
1. **Strict Penalties for Violations:**
 - Fine up to **₹1 lakh** or **imprisonment up to 5 years** for polluters.
 - Continued violations can result in **higher penalties**.
1. **Environmental Standards & Monitoring:**
 - Establishes **air and water quality standards**.
 - Provides for **periodic inspection and reporting**.
1. **Handling of Hazardous Substances:**
 - Regulates the **storage, handling, and disposal** of toxic chemicals.

- Example: **Control over industrial waste and radioactive materials.**
1. **Public Participation & Awareness:**
 - Encourages **citizen involvement** in environmental protection.
 - Example: **Public interest litigation (PIL)** allows citizens to report environmental violations.
 1. **Legal Authority & Enforcement:**
 - Empowers **courts to take action** against environmental offenders.
 - Provides for the establishment of **environmental courts** if needed.

Conclusion:

The **Environmental Protection Act, 1986** is a **comprehensive law** that gives the government the **power to regulate pollution, enforce penalties, and ensure environmental sustainability**. It remains one of the **strongest legal tools** for environmental protection in India.

Methods of Disposal of Municipal Solid Waste Management (8 Marks)

Municipal Solid Waste (MSW) includes household waste, commercial waste, and construction debris. Proper disposal is essential for **public health and environmental sustainability**.

1. Landfilling

- **Process:** Waste is dumped in **designated landfill sites** and covered with soil.
- **Types:**
 - **Sanitary Landfills** – Lined with impermeable layers to prevent pollution.
 - **Open Dumping** – Unregulated, leads to pollution (not recommended).
- **Advantages:**
 - Cost-effective and easy to implement.
- **Disadvantages:**
 - Produces **methane gas**, causing air pollution.
 - Can contaminate **groundwater** if not managed properly.

2. Incineration (Waste-to-Energy)

- **Process:** Waste is **burned at high temperatures** in incinerators, converting it into energy.
- **Advantages:**
 - Reduces waste volume by **90%**.
 - Generates **electricity** from waste heat.
- **Disadvantages:**
 - Releases **toxic gases (dioxins, furans)** if not properly controlled.

- High **operational costs**.

3. Composting (Biological Treatment)

- **Process:** Organic waste (food, garden waste) is decomposed by **microorganisms** into compost.
- **Types:**
- **Aerobic Composting** – Requires oxygen, produces quality compost.
- **Vermicomposting** – Uses earthworms to break down waste.
- **Advantages:**
- Produces **natural fertilizer** for agriculture.
- Reduces landfill waste.
- **Disadvantages:**
- Slow process, requires **large space**.

4. Recycling and Reuse

- **Process:** Waste materials (plastic, paper, metal, glass) are processed for **reuse**.
- **Steps:**
- Collection → Sorting → Cleaning → Processing into new products.
- **Advantages:**
- Saves **natural resources**.
- Reduces **pollution and landfill waste**.
- **Disadvantages:**
- Requires **public awareness and infrastructure**.

5. Pyrolysis (Plasma Gasification)

- **Process:** Waste is heated in the absence of oxygen to produce **syngas and bio-oil**.
- **Advantages:**
- Converts waste into **fuel and electricity**.
- Reduces **greenhouse gas emissions**.
- **Disadvantages:**
- High **initial investment cost**.

Conclusion:

Effective **solid waste management** requires a **combination** of methods like **landfilling, recycling, composting, and energy recovery**. Governments should promote **waste segregation and public awareness** to ensure a **sustainable waste management system**.

Principle and Various Steps Involved in Energy Management (8 Marks)

Principle of Energy Management:

Energy management refers to the efficient use of energy resources to minimize waste and maximize productivity. The **key principles** are:

1. **Conservation First** – Reduce energy waste before investing in new energy sources.
2. **Use of Renewable Energy** – Shift from fossil fuels to solar, wind, and hydro energy.
3. **Energy Efficiency** – Use energy-efficient appliances and industrial processes.
4. **Monitoring and Control** – Implement automated systems for real-time energy tracking.
5. **Sustainability** – Balance economic growth with environmental protection.

Steps Involved in Energy Management:

1. **Energy Audit:**
 - Identify **energy consumption patterns** in industries, buildings, or homes.
 - Conduct **measurements and analysis** of energy use.
1. **Setting Energy Goals:**
 - Define **specific targets** for energy savings (e.g., reduce electricity use by 10%).
 - Develop a **roadmap** for achieving these goals.
1. **Implementation of Energy Efficiency Measures:**
 - Use **LED lighting** instead of incandescent bulbs.
 - Upgrade to **energy-efficient appliances and machinery**.
 - Optimize **HVAC (Heating, Ventilation, and Air Conditioning)** systems.
1. **Use of Renewable Energy Sources:**
 - Integrate **solar panels, wind turbines, and biomass energy** in operations.
 - Promote **grid-connected and off-grid renewable systems**.
1. **Employee Training and Awareness:**
 - Educate employees on **energy conservation practices**.
 - Conduct **workshops on energy efficiency**.
1. **Monitoring and Continuous Improvement:**
 - Use **smart meters and IoT-based energy monitoring systems**.
 - Regularly update energy management plans for continuous efficiency.
1. **Government Policies and Compliance:**
 - Follow **energy regulations** like the **Energy Conservation Act, 2001 (India)**.
 - Avail **subsidies and incentives** for energy-efficient practices.

Conclusion:

Effective **energy management** ensures **cost savings**, **reduces environmental impact**, and promotes **sustainability**. By **following these steps**, industries and households can **reduce their carbon footprint** while improving efficiency.

OR

8(b) Write a Note on:

(i) *Tidal Energy*

- **Definition:** Tidal energy is a form of renewable energy harnessed from the movement of ocean tides.
- **Working Principle:**
- Uses **tidal barrages, tidal stream generators, or dynamic tidal power** to convert **kinetic energy of tides** into electricity.
- **Advantages:**
- Predictable and reliable (unlike solar/wind).
- No greenhouse gas emissions.
- **Disadvantages:**
- High initial setup cost.
- Can impact marine ecosystems.

(ii) *OTE (Ocean Thermal Energy)*

- **Definition:** Ocean Thermal Energy (OTE) utilizes the temperature difference between warm surface water and cold deep water to generate electricity.
- **Working Principle:**
- A **heat engine** uses the warm water to vaporize a working fluid (like ammonia).
- The vapor drives a **turbine**, generating electricity.
- Cold deep water then **condenses** the vapor back into liquid.
- **Advantages:**
- Provides **continuous** energy.
- Uses **natural temperature gradients**.
- **Disadvantages:**
- Works best in tropical regions with **large temperature differences**.
- High cost of underwater infrastructure.

(iii) *GTE (Geothermal Energy)*

- **Definition:** Geothermal Energy (GTE) is heat energy generated from the Earth's interior, used for power generation and direct heating.

- **Working Principle:**
- Hot **magma** beneath the Earth heats underground water, creating **steam**.
- Steam drives **turbines**, producing electricity.
- Used in **geothermal power plants** and direct heating applications.
- **Advantages:**
- Highly sustainable and available **24/7**.
- Low environmental impact compared to fossil fuels.
- **Disadvantages:**
- Limited to **geothermal hotspots** (e.g., Iceland, Japan).
- Drilling deep wells is **expensive** and may cause **seismic activity**.

Conclusion:

Tidal Energy, OTE, and GTE are **promising renewable energy sources** with unique benefits and challenges. With **advancements in technology**, they can play a major role in sustainable energy production.